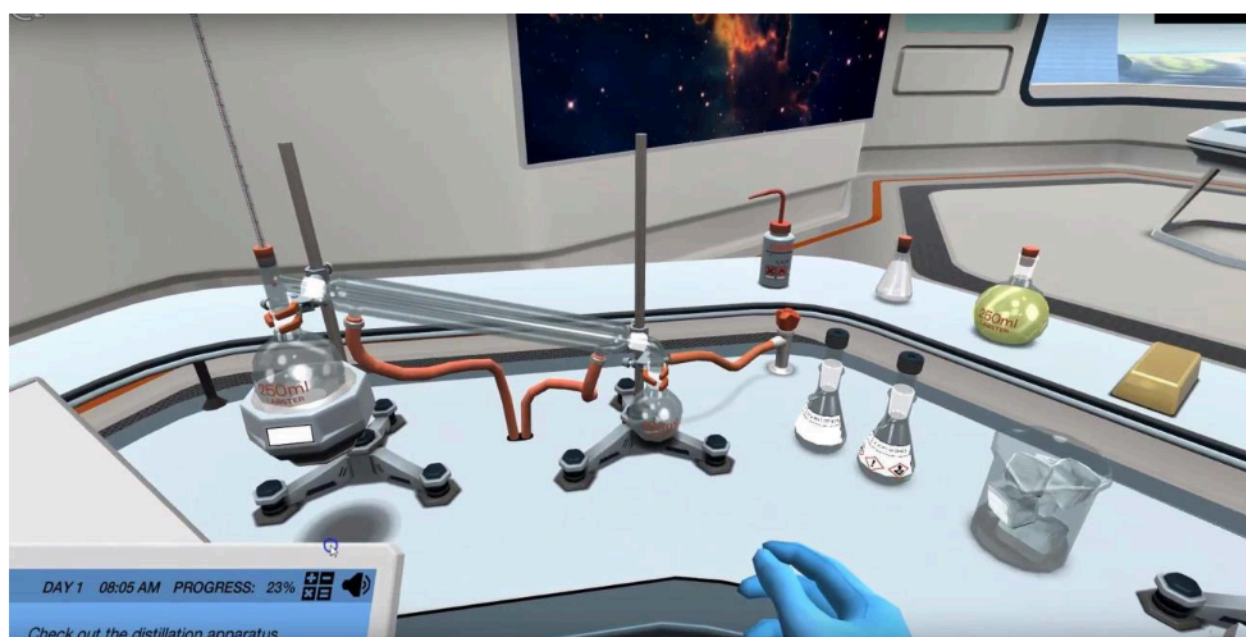


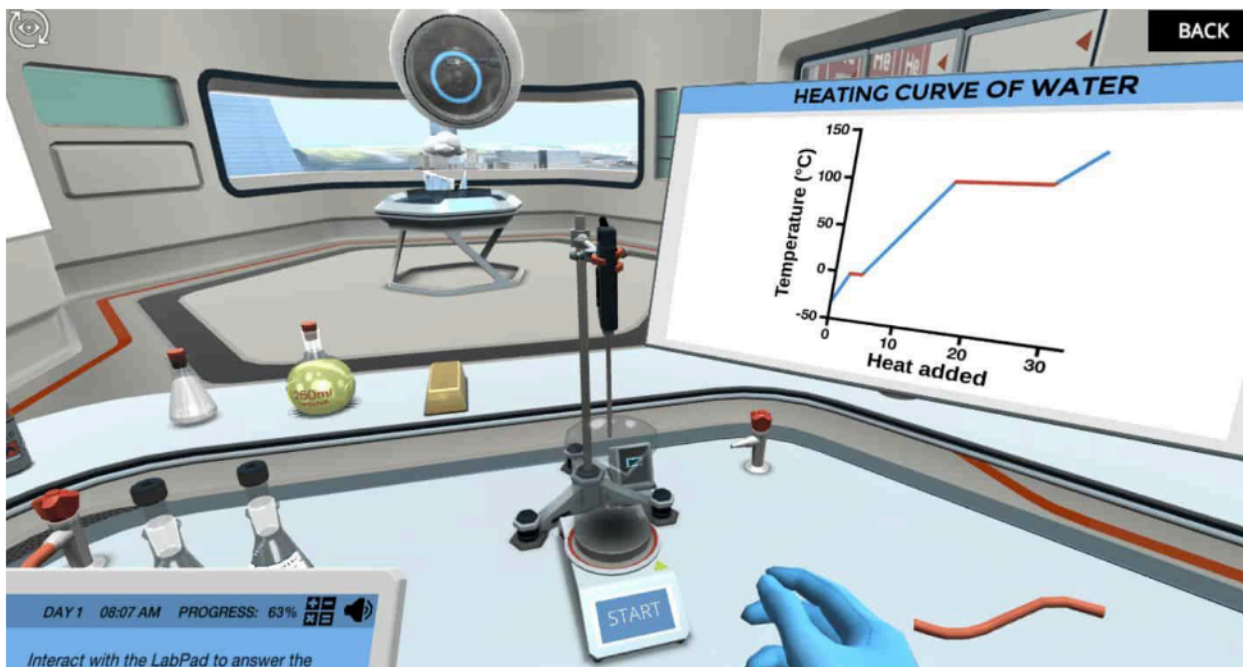
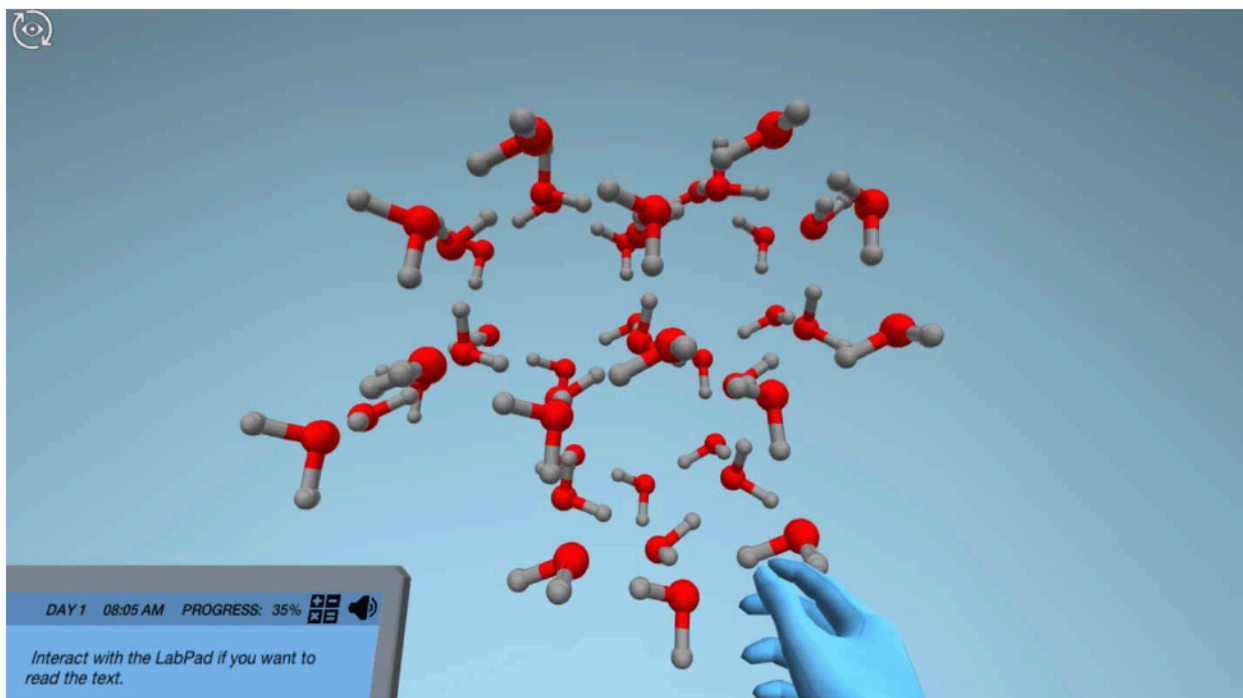
Matter and Phase Changes: Distil ethanol

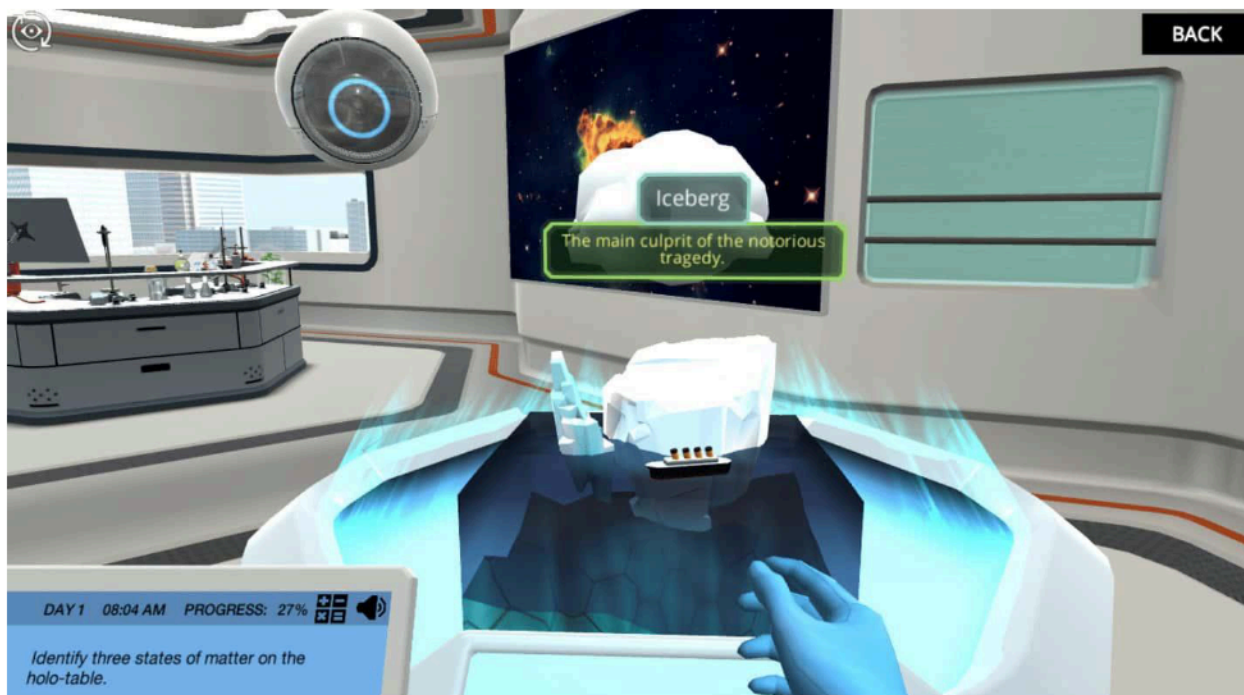
In this assignment, you will be playing an immersive online learning simulation.

Below you can read the main learning outcomes and techniques you'll practice, as well as a brief introduction and storyline to the simulation.

Once you are ready, you can start the simulation by pressing the simulation launch button/link.







Learning Objectives

At the end of this simulation, you will be able to...

- Explain solid, liquid, and gas states in terms of particle interaction and bonding energy
- Describe and explain the characteristics of a phase change
- Read a phase diagram and explain each region
- Interpret a heating curve of a given substance
- Explain the difference between a heating curve and a phase diagram, and the difference between specific heat and latent heat
- Understand the basic steps of performing a distillation
- Distinguish between physical and chemical properties of matter and classify changes of matter as physical or chemical
- Name the main phase changes: boiling, evaporation, freezing, melting, and sublimation

Techniques:

- Distillation

About this Learning Simulation

Get ready to encounter all three states of matter. In this simulation, you will manipulate the states of matter and inspect the phase changes. Will you be able to see that almost everything you interact with matters?

Explore the intermolecular forces in matter

In the Matter and Phase Changes simulation, you will meet Jo who has a limited fuel supply and therefore wants to produce bioethanol. In order to help her, you must first master the definition of matter and the process of phase changes. You will dive into the water and examine the dynamics of intermolecular forces of water molecules during the phase changes. With the help of 3D animations, you will learn about the molecular configuration in three states of matter: Solid, liquid, and gas.

Generate heating curves

Once you've mastered the basics, you are ready to learn about additional aspects of phases changes: Specific heat, latent heat, and pressure. You will utilize a temperature probe and a heat plate to generate a heating curve and then compare the heating curve of water, ethanol, and metal. By doing so, you will be able to decode the meaning of each line of the heating curves.

Perform distillation

At the end of the simulation, you will put your newly acquired knowledge to the test!. You are going to use the distillation technique to achieve ethanol with higher purity.

Will you be able to produce high purity ethanol for Jo?